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- Find shortest cycle through all vertices
- Find optimal coloring
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- Linear programming: class of optimization problems where constraints and optimization criterion are linear functions

Avoiding Scurvy

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  grapefruit costs $1;
  we have budget of $2/day

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- Orange has 50gm of vitamin C,
  grapefruit has 75gm of vitamin C,
  maximize daily vitamin C intake.

Avoiding Scurvy. Plot

$$\max 2x + 3y$$

$$x + y \leq 2$$
$$x + 2y \leq 3$$
$$x \geq 0$$
$$y \geq 0$$

Avoiding Scurvy. Plot

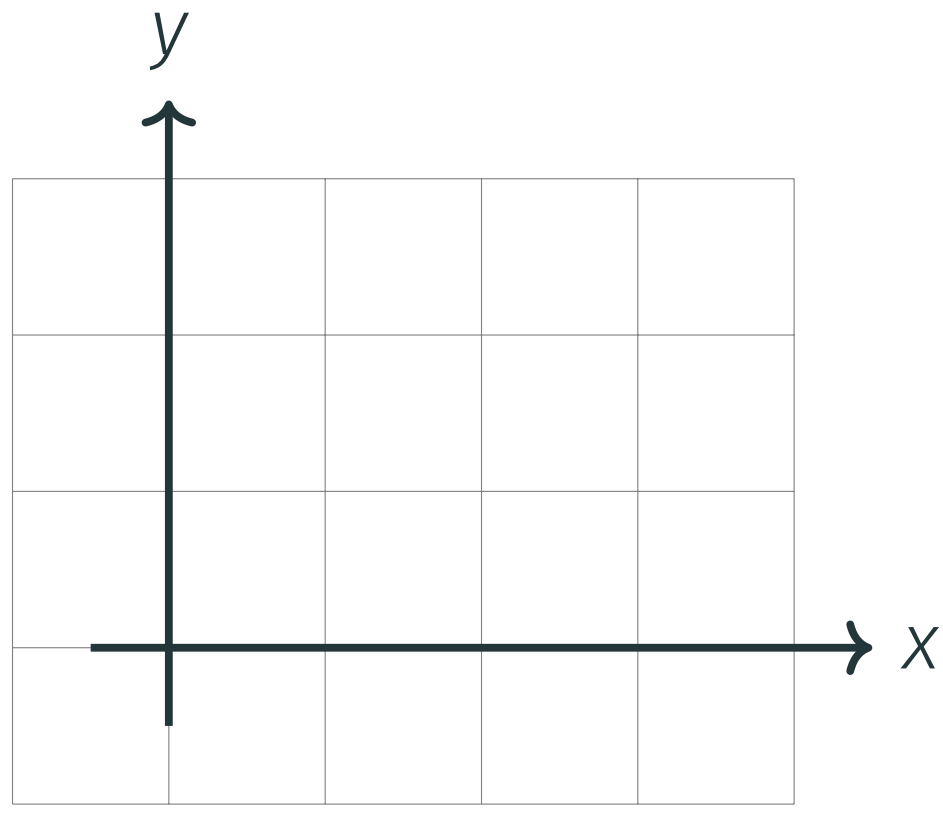
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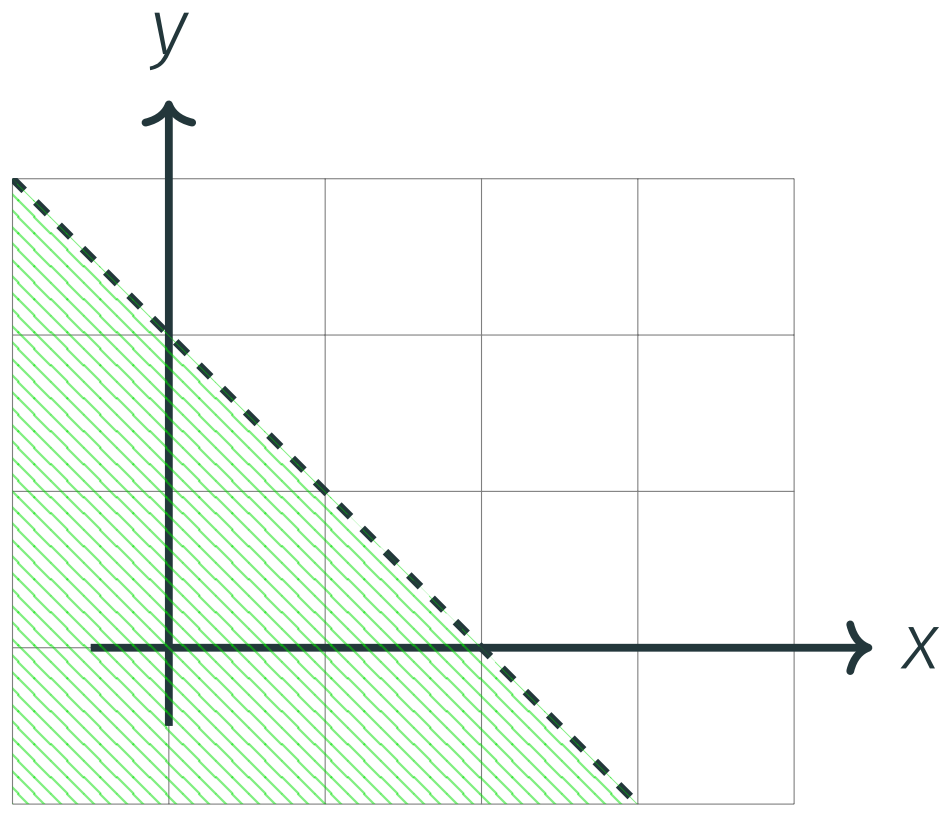
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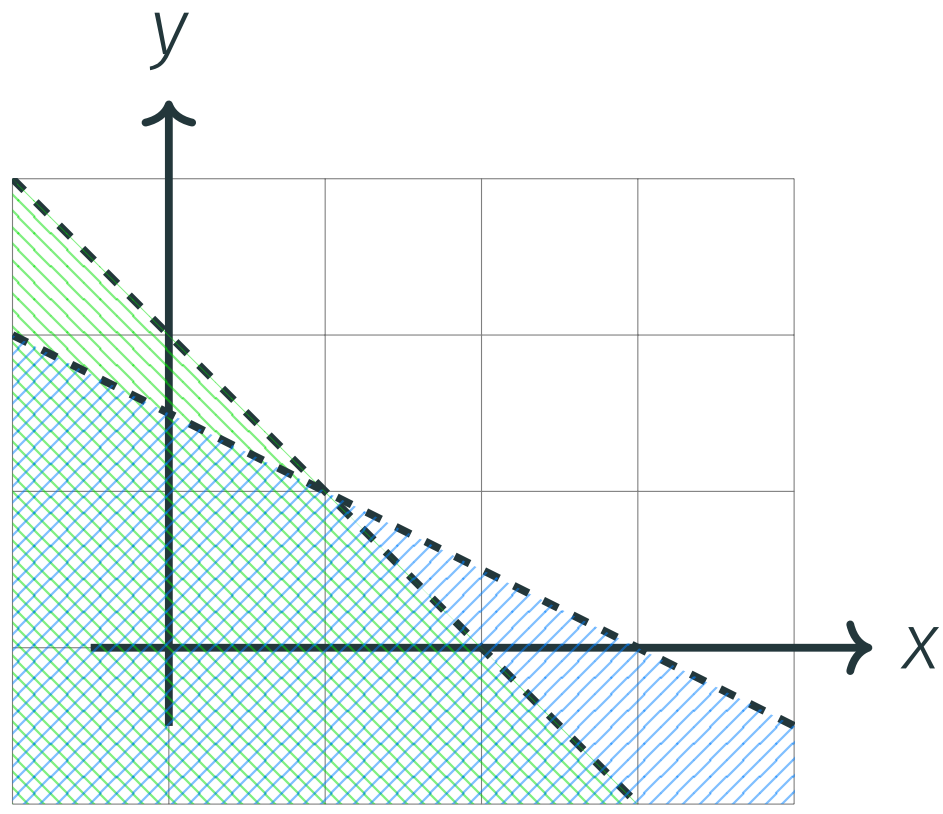
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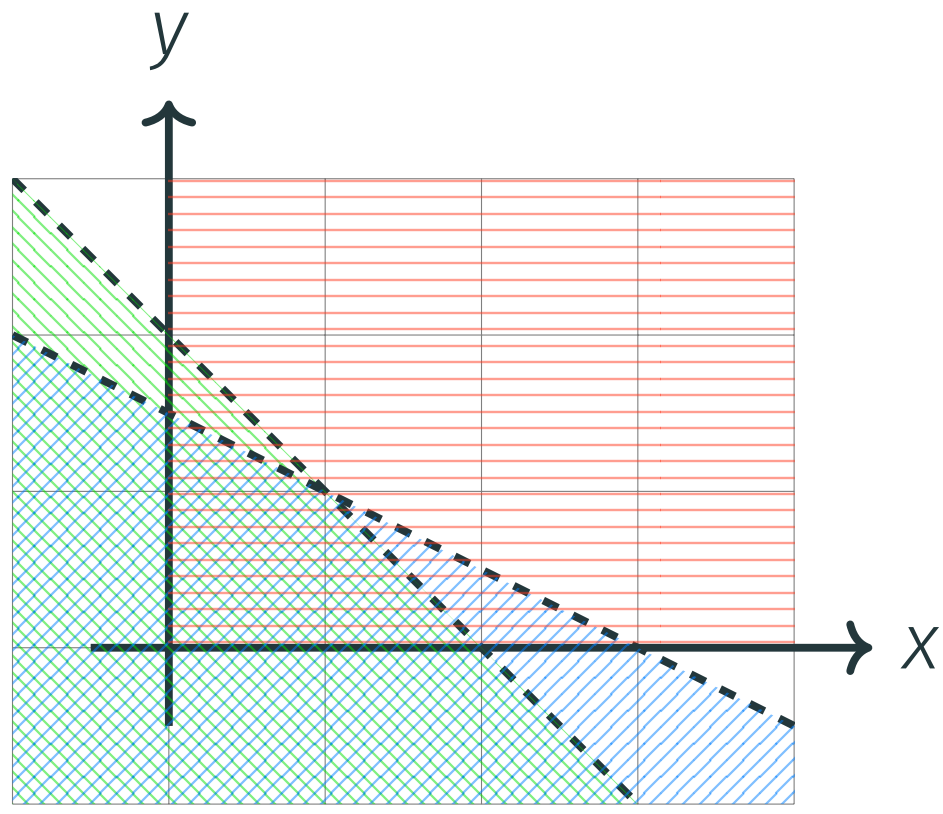
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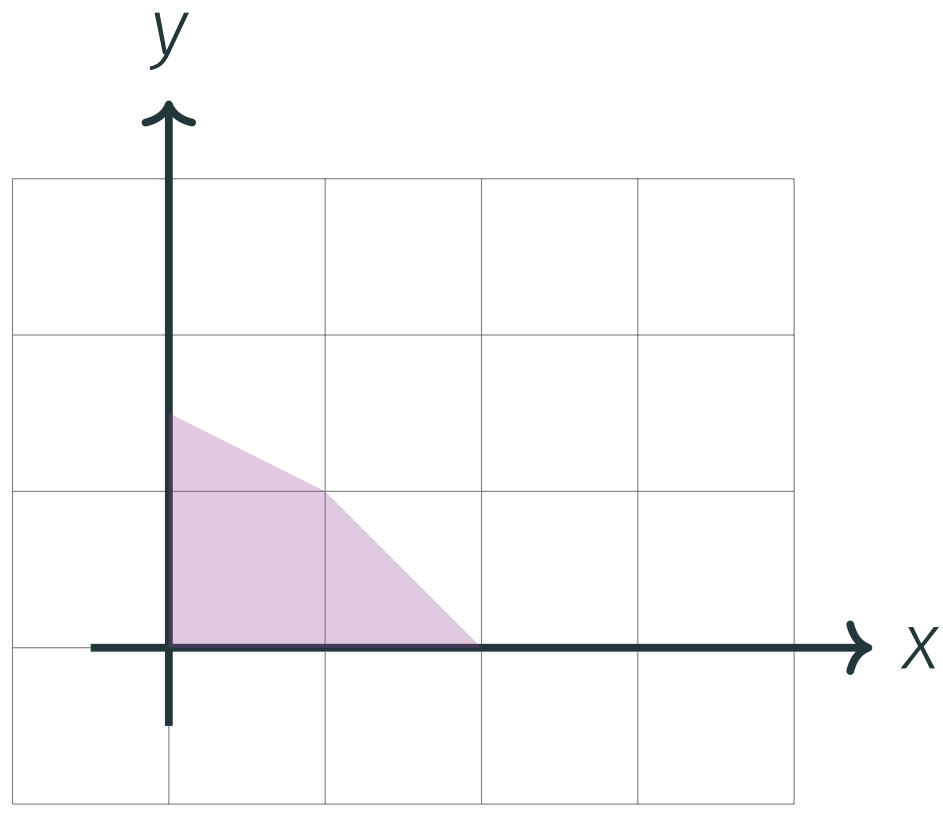
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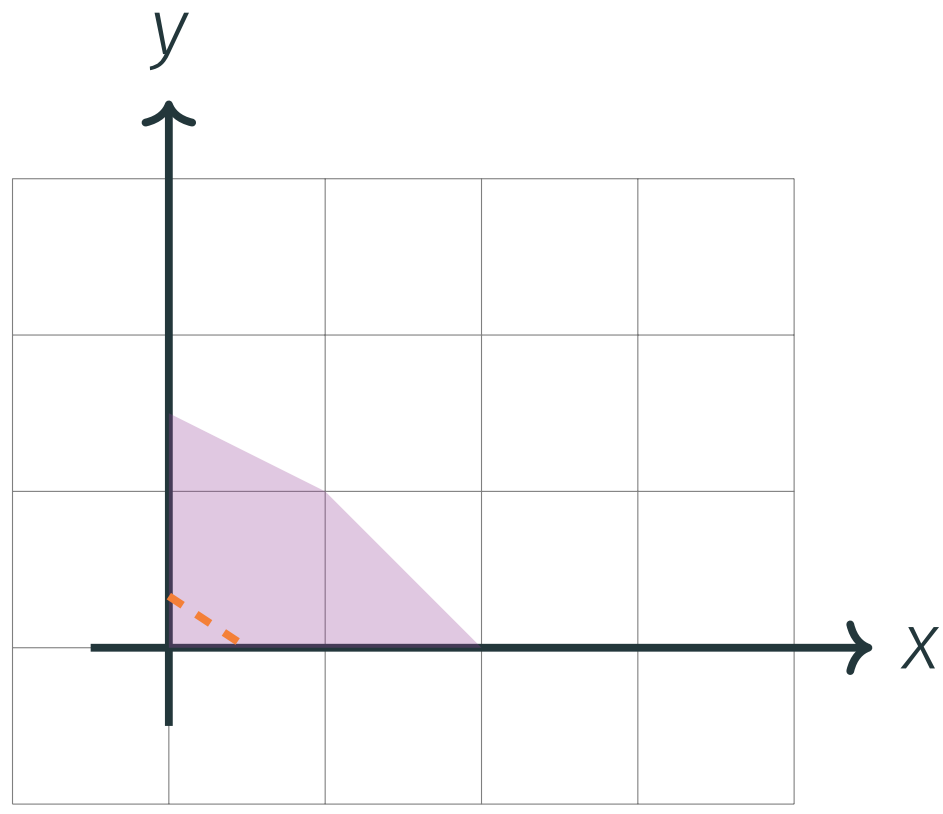
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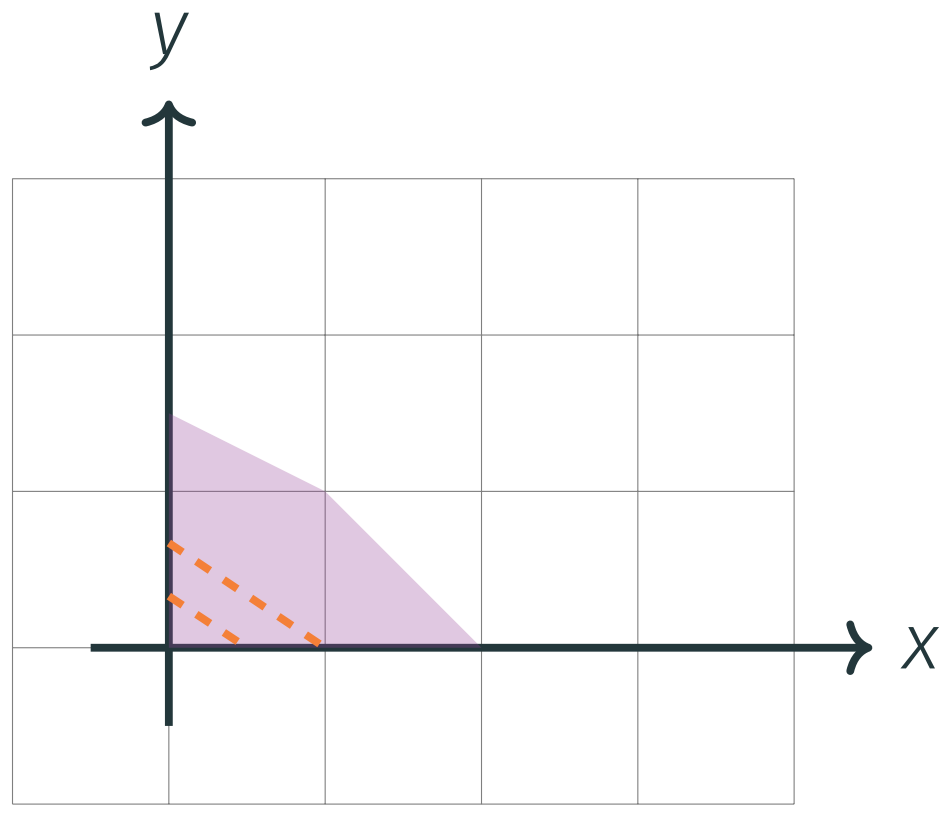
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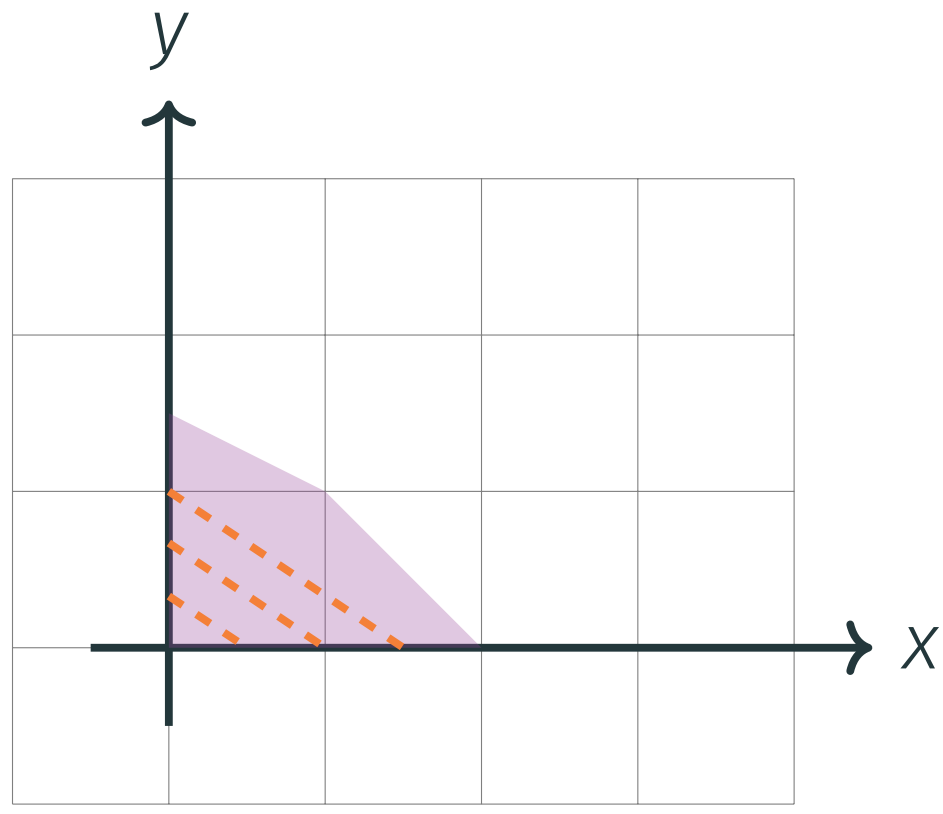
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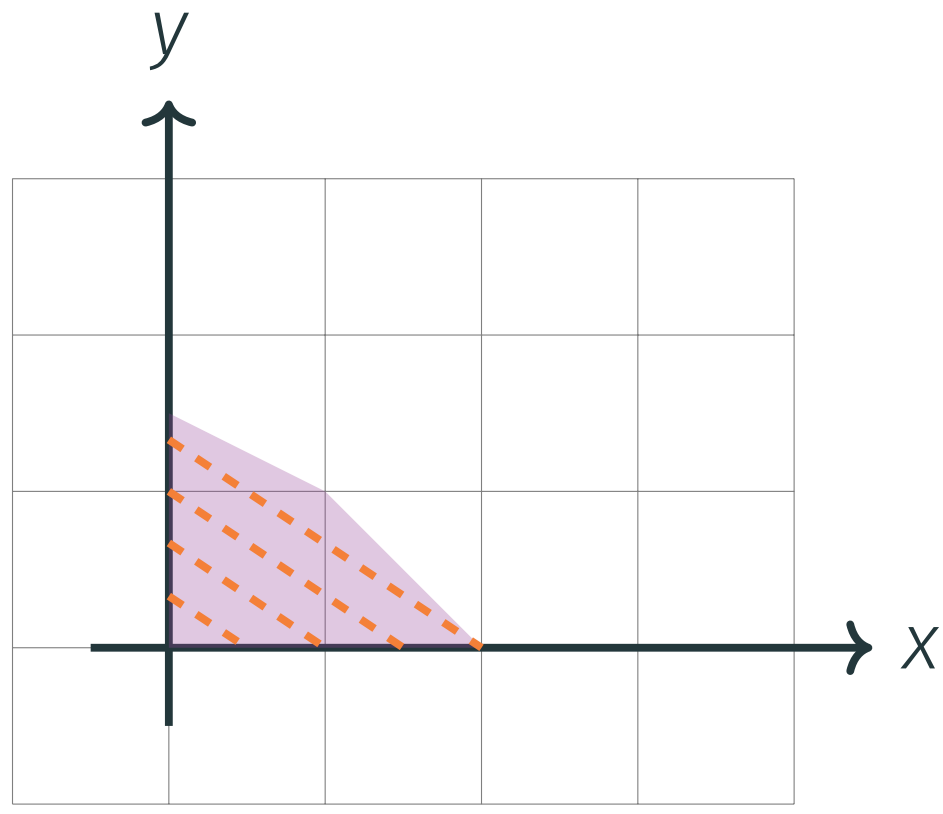
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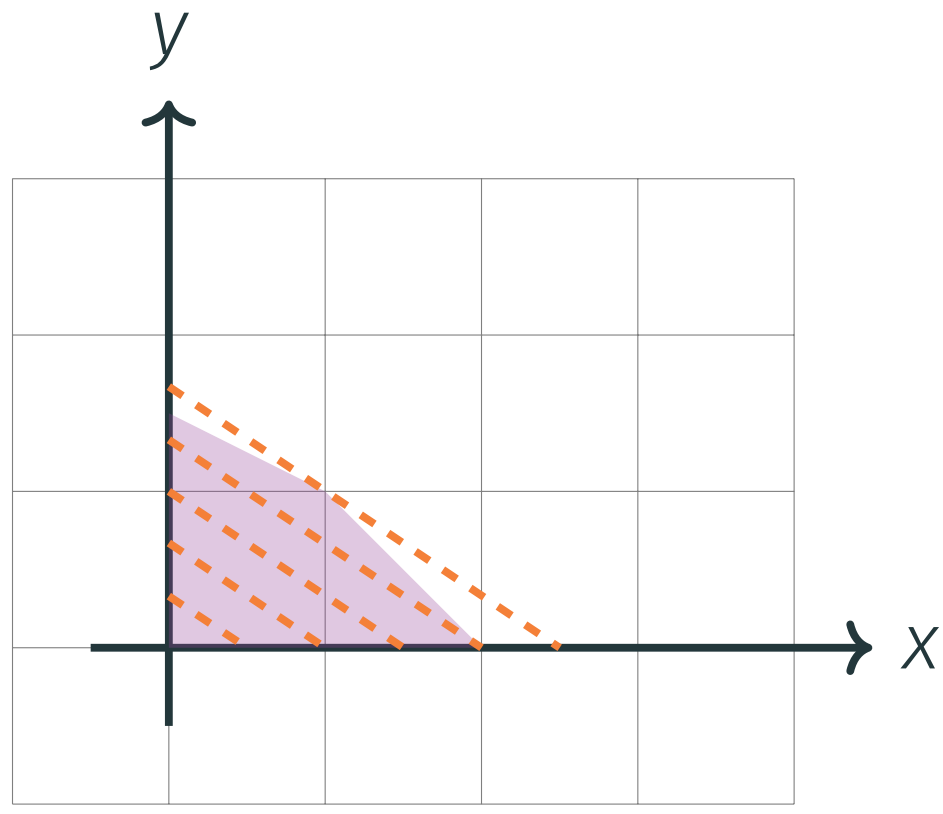
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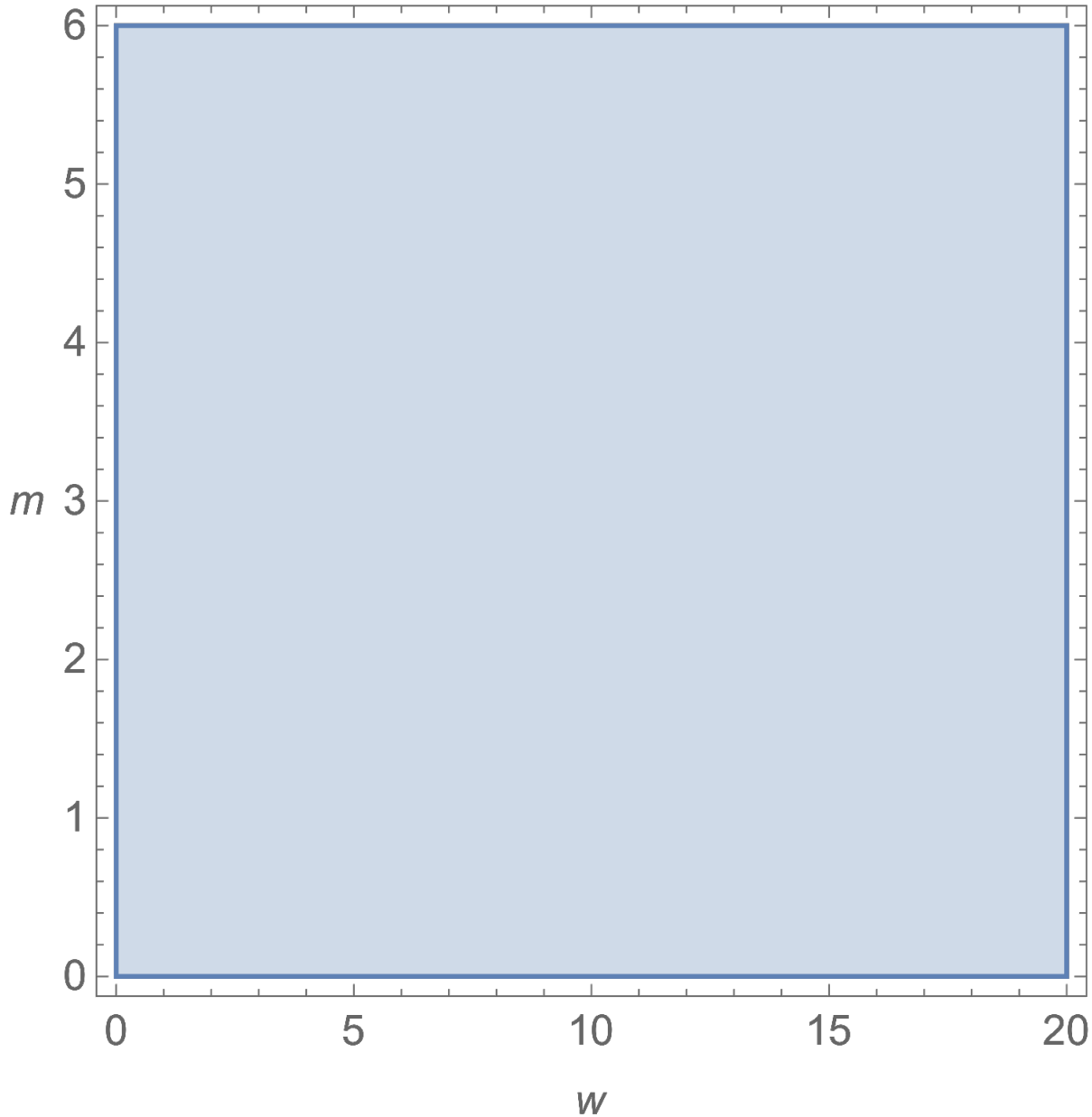
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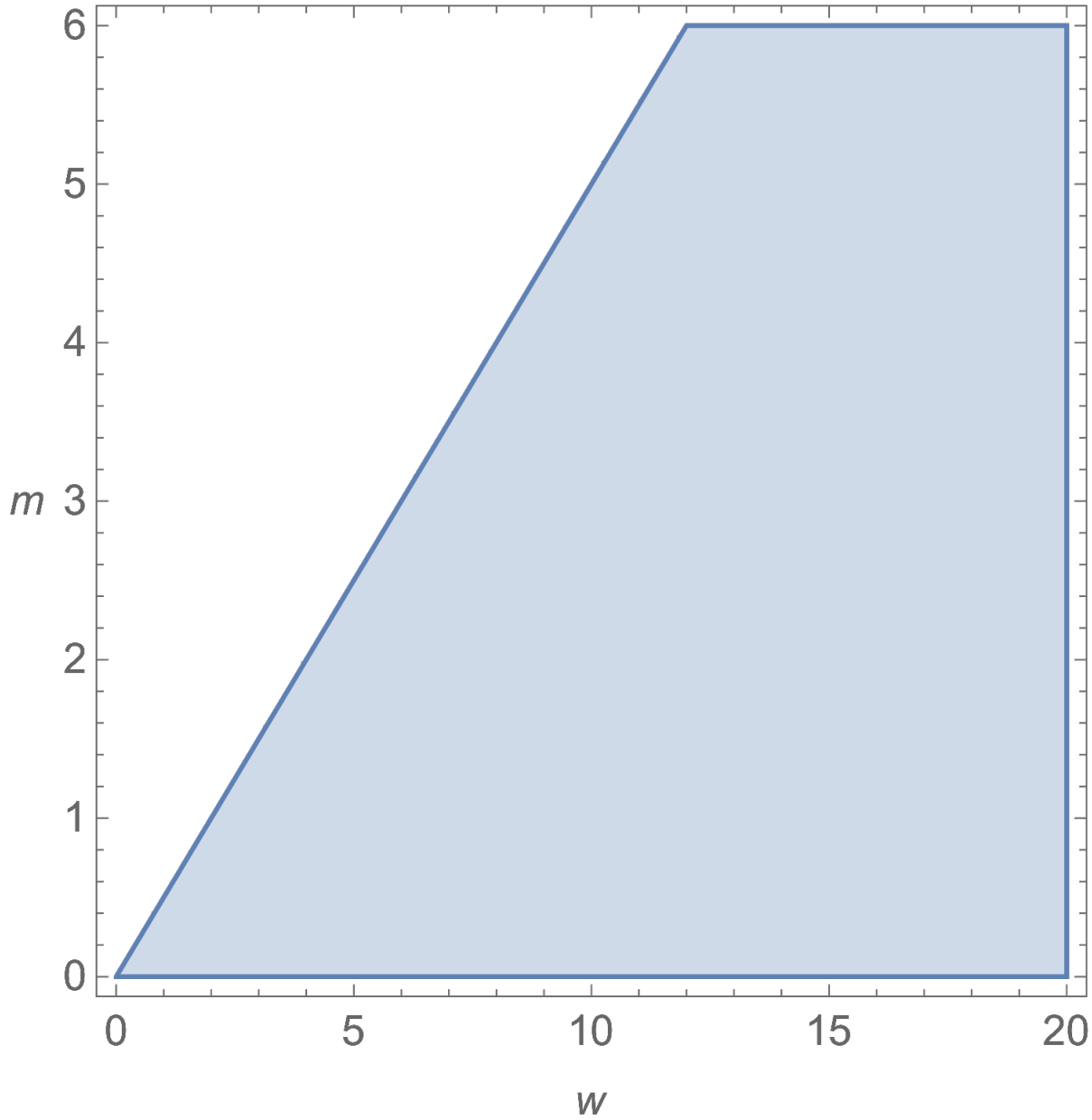
Profit Maximization

- We have 6 machines and 20 workers
- A machine takes two workers to operate
- Each machine produces 10 chocolates/hour, each worker produces 5 chocolates/hour
- We need to produce at most 100 chocolates/hour
- Each chocolate costs $10, each worker gets $40 per hour

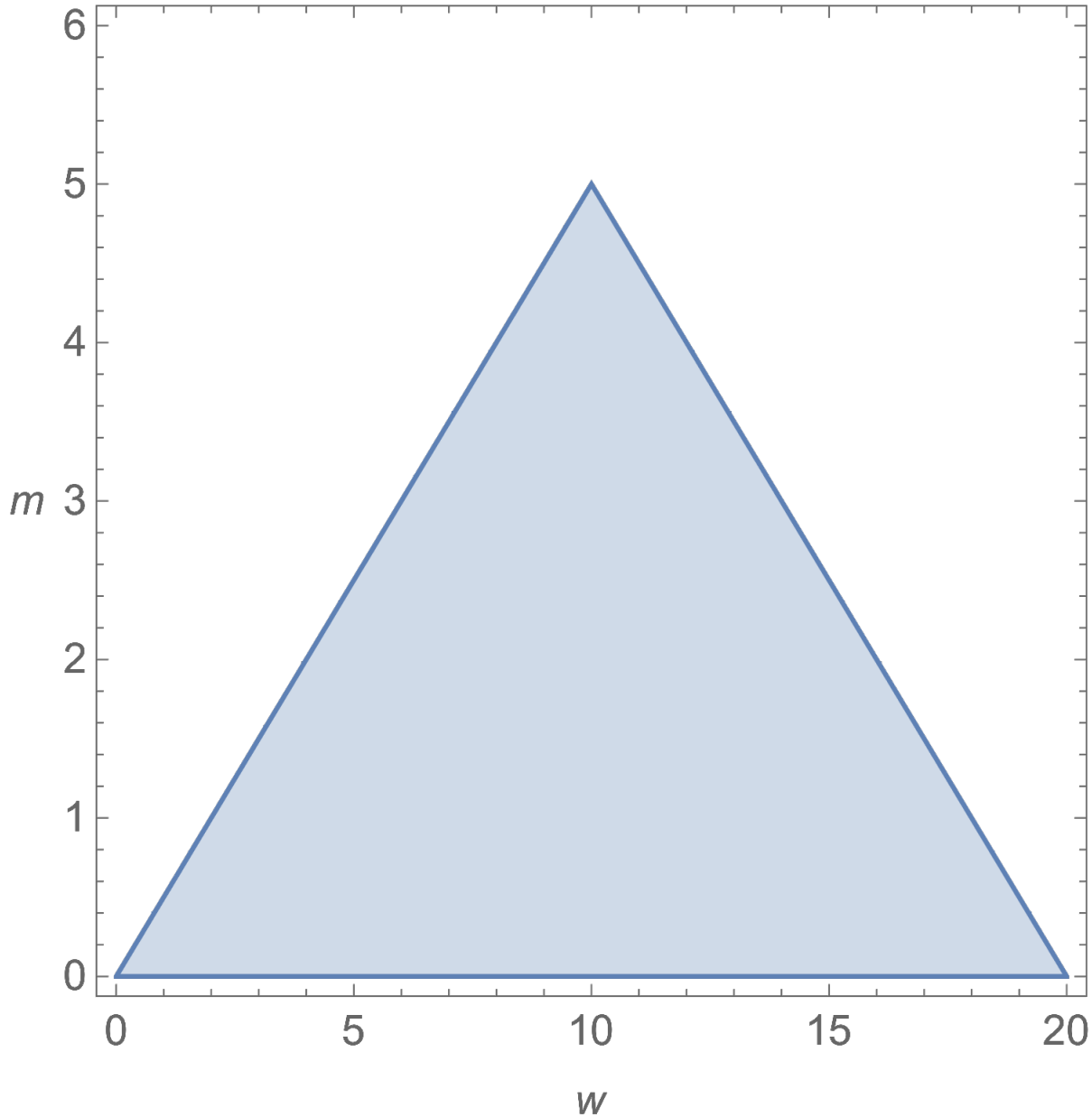
Workers and Machines



Two Workers Operate a Machine



Chocolate Demand



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  - $h(a_1, \ldots, a_d) < 0$ for all spam emails
  - $h(a_1, \ldots, a_d) > 0$ for all ham emails

Linear Programming

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- Find real numbers $x_1, \ldots, x_n$ that satisfy linear constraints

$$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \geq b_1$$
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- So that linear objective is maximized

$$c_1x_1 + c_2x_2 + \ldots + c_nx_n$$

Equivalent Formulations

- Turn minimization problem into maximization problem:

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Input is a matrix $A \in \mathbb{R}^{m \times n}$, and vectors $b \in \mathbb{R}^m$ and $c \in \mathbb{R}^n$

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$$\text{maximize } cx = \begin{bmatrix} c_1 & \dots & c_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = c_1x_1 + \dots c_nx_n$$

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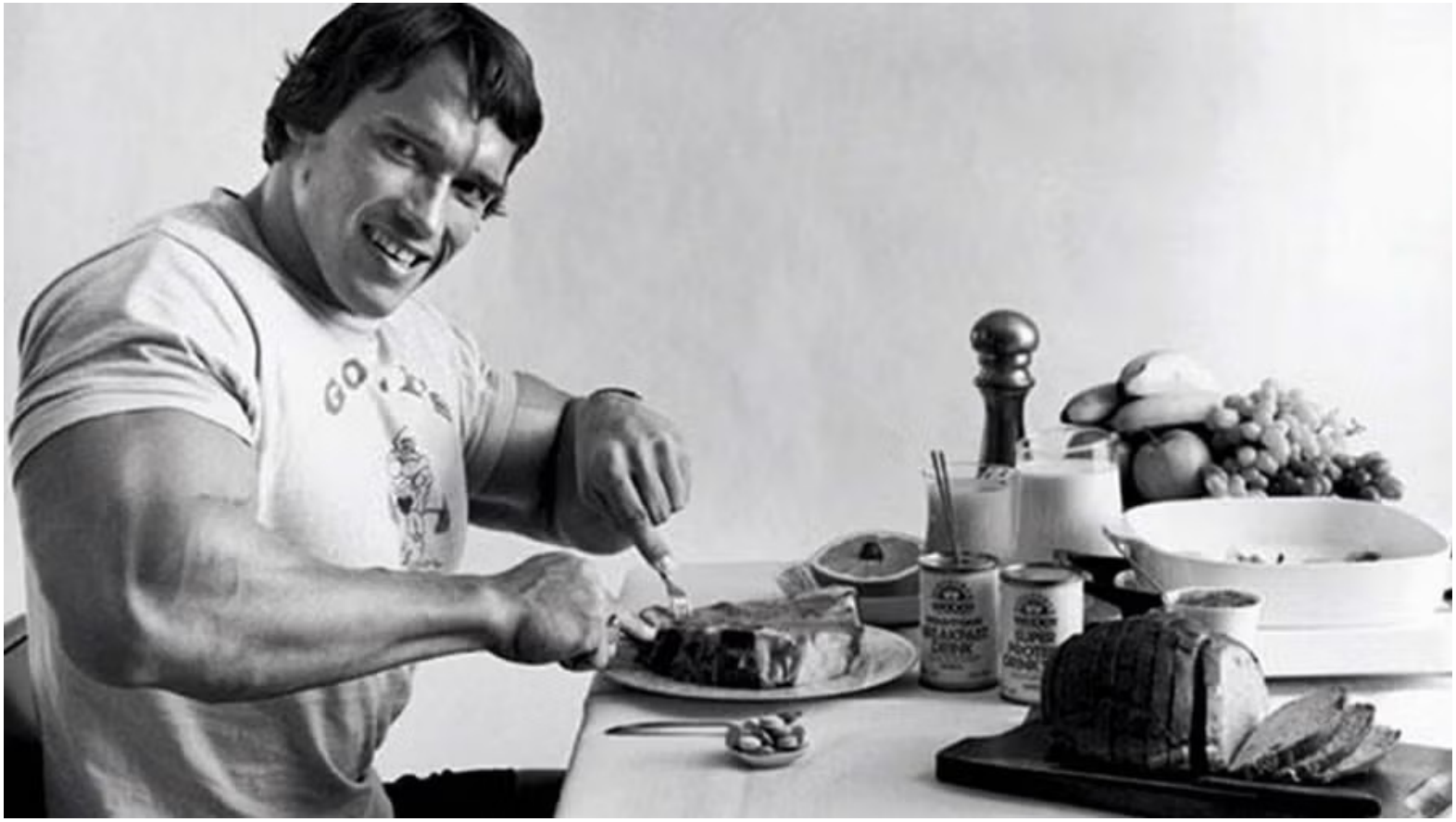
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- Dantzig's algorithm is "One of top 10 algorithms of the 20th century"

LP DUALITY

Plan your diet (ugh, not again)

- you need at least 12g of protein and 8g of fiber
- minimize cost



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I don’t believe you!



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Simplex Method

Theorem

A linear function takes its maximum and minimum values on vertices

- Start at any vertex
- While there is an adjacent vertex with higher profit
    - Move to that vertex